

Lecture 34 — Embedded Real-Time Operating Systems

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Placeholder

This lecture has not yet been written.

The intended sections are:

1. What an RTOS is — and is not
2. Life without an MMU: flat address spaces and the MPU
3. Static allocation and why `malloc` is suspect
4. FreeRTOS anatomy: tasks, the tick, and the scheduler
5. Priority-based preemptive scheduling in practice
6. Synchronization: queues, semaphores, priority inheritance
7. Interrupts: deferring work out of the ISR
8. Tickless idle and low-power operation
9. Zephyr as the modern alternative
10. Choosing: embedded RTOS vs. real-time Linux

References